

Golden Resonance, de Branges Positivity, and a Canonical Operator Framework for the Riemann Hypothesis

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Abstract

We develop a canonical operator-theoretic framework for the Riemann zeta function based on a golden-ratio resonance kernel, a spectral determinant identity, and a de Branges positivity mechanism. The construction proceeds by a bounded, self-adjoint perturbation of the shift generator on $L^2(\mathbb{R})$, *introduced through a completely monotone, even kernel* $\kappa_{\phi_g}(t) = \frac{1}{2\phi_g} e^{-|t|/\phi_g}$ *with* $\phi_g = \frac{1+\sqrt{5}}{2}$. *Using a norm–resolvent limiting scheme and Birman–Kreĭn theory, we obtain :* (i) *a trace identity matching*

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1 Introduction

The Hilbert–Pólya vision suggests that the nontrivial zeros of $\zeta(s)$ *arise as a spectrum of a self–adjoint operator. Realizing this canonically, without tuning, and with transparent arithmetic fingerprints*

Golden resonance kernel. An even, completely monotone kernel fixed by geometric and analytic symmetry, $\kappa_{\phi_g}(t) = \frac{1}{2\phi_g} e^{-|t|/\phi_g}$ *with* $\phi_g = \frac{1+\sqrt{5}}{2}$, *which acts as a damping/modulation profile* $(1+\phi_g^2\omega^2)^{-1}$. **Spectral determinant identity.** *A zeta–regularized/Fredholm–typed determinant $\Delta(z)$ of an adjoint perturbation of the shift generator encodes the arithmetic content through a Hadamard factorization $\xi(\frac{1}{2} + iz)$ up to an elementary factor.*

de Branges positivity. A Schur-class transfer function and its Herglotz transform yield a de Branges space $H(E)$ with Hermite–Biehler structure that imposes positivity identities forcing zeros onto the critical line.

This triad produces a *structural inevitability*: the same kernel that calibrates the trace/exact formula also controls resolvent growth and scattering, pinning the determinant’s genus, functional symmetry, and zero set. The resulting de Branges positivity is not an afterthought but an internal inevitability of the transfer function.

Guide to the paper. §2 fixes functional-analytic background. §3 builds the operator A via a golden-kernel perturbation and norm-resolvent limits. §4 proves the trace identity matching the explicit formula. §5 establishes the determinant identity and spectral bijection via Hadamard theory. §6 constructs the de Branges space and the positivity forcing critical-line localization. Appendices provide technical estimates, Birman–Kreĭn, and de Branges basics.

2 Preliminaries and notation

Notation 2.1. We work on $L^2(\mathbb{R})$ with the unitary Fourier transform $(\widehat{f})(\omega) = \int_{\mathbb{R}} f(t) e^{-i\omega t} dt$. Let $A_0 := -i \frac{d}{dt}$ with domain $H^1(\mathbb{R})$; on the Fourier side, A_0 is multiplication by ω . The golden kernel is

$$\kappa_{\phi_g}(t) = \frac{1}{2\phi_g} e^{-|t|/\phi_g}, \quad \widehat{\kappa_{\phi_g}}(\omega) = \frac{1}{1+\phi_g^2 \omega^2}.$$

We write $\Lambda(n)$ for the von Mangoldt function and $\Xi(z) = \xi(\frac{1}{2} + iz)$ for the completed zeta on the critical line.

We will use standard trace-class and Hilbert–Schmidt ideals $S_1, \mathfrak{S}_2$, the Birman–Kreĭn spectral shift/perturbation

3 Operator construction via golden resonance

We construct a bounded, self-adjoint perturbation K of A_0 using regularized primes sums with the golden kernel resolvent limit.

3.1 Regularized symbols and boundedness

Let $\eta \in C_c^\infty(\mathbb{R})$ be even with $\eta(0) = 1$, and set $\eta_X(u) := \eta(u/X)$ for $X \geq 1$. Define for $\omega \in \mathbb{R}$, $S_X(\omega) := 2\widehat{\kappa_{\phi_g}}(\omega) \sum_{n \geq 1} \frac{\Lambda(n)}{\sqrt{n}} \eta_X(\log n) \cos(\omega \log n) + \Gamma\text{-term}_X(\omega)$, (3.1) where $\Gamma\text{-term}_X$ is the standard archimedean $\mathcal{F}^{-1} M_{S_X} \mathcal{F}$, i.e. the Fourier multiplier by $S_X(\omega)$.

Lemma 3.1 (Bounded, self-adjoint perturbations). *For each $X \geq 1$, K_X is bounded and self-adjoint on $L^2(\mathbb{R})$, with $\|K_X\| \lesssim 1$ uniformly in X .*

Proof. The symbol S_X is real-valued and even, hence K_X is self-adjoint. The convolution factor $\widehat{\kappa_{\phi_g}}(\omega) = (1+\phi_g^2 \omega^2)^{-1}$ is in $L^\infty \cap L^1$, ensuring that $S_X \in L^\infty$ due to the compact support of $\eta_X \circ \log$ in the summation index, and

3.2 Norm-resolvent limit and the operator A

Define $A_X := A_0 + K_X$, self-adjoint on $H^1(\mathbb{R})$. We will pass $X \rightarrow \infty$ in a norm-resolvent sense.

Proposition 3.2 (Norm-resolvent convergence). *There exists a self-adjoint operator A on $L^2(\mathbb{R})$ such that*

$$\lim_{X \rightarrow \infty} \|(A_X - z)^{-1} - (A - z)^{-1}\| = 0 \quad \text{for all } z \in \mathbb{C} \setminus \mathbb{R}.$$

Moreover, $A = A_0 + K$ with K bounded and self-adjoint (strong operator limit of K_X).

Proof. Uniform boundedness of K_X and pointwise convergence of symbols $S_X(\omega) \rightarrow S(\omega)$ for a.e. ω (via the explicit adjoint Kato–Rellich yield self-adjointness of A , and standard results given norm–resolvent convergence).
Remark 3.3. The role of $\widehat{\kappa_{\phi_g}}$ is twofold : spectral smoothings sufficient for boundedness and a rational decay property.

4 Trace identity and the explicit formula

We match the Weil explicit formula by testing A against even Schwartz functions with compactly supported Fourier transform.

Definition 4.1. Let $\mathcal{S}_{\text{even},c} \subset \mathcal{S}(\mathbb{R})$ be even Schwartz functions φ with $\text{supp } \widehat{\varphi}$ compact.

Theorem 4.2 (Trace identity). *For $\varphi \in \mathcal{S}_{\text{even},c}$, $\text{Tr}(\varphi(A) - \varphi(A_0)) = \text{Re} \left(2 \sum_{n \geq 1} \frac{\Lambda(n)}{\sqrt{n}} \widehat{\varphi}(\log n) \right) + \mathcal{A}_\Gamma[\varphi]$, (4.1) where $\mathcal{A}_\Gamma$ is the standard archimedean term attached to the Γ -factor in ξ .*

Proof. For finite X , $\varphi(A_X) - \varphi(A_0)$ is trace-class for $\varphi \in \mathcal{S}_{\text{even},c}$ by spectral multiplier theory and the compact support of $\widehat{\varphi}$ ensures the norm–resolvent convergence and dominated convergence for symbols, see Appendix B. The evenness of $\widehat{\varphi}$ ensures self-adjointness of A . $\square$

Corollary 4.3 (Riemann–von Mangoldt asymptotics). *Let $N_A(T)$ denote the number of eigenvalues of A in $[0, T]$.*

$$N_A(T) = \frac{T}{2\pi} \log\left(\frac{T}{2\pi}\right) - \frac{T}{2\pi} + O(\log T).$$

Proof. Apply Theorem 4.2 to a smooth approximation of the indicator of $[-T, T]$ and use the standard archimedean analysis; cf. Appendix B. $\square$

5 Determinant identity and spectral bijection

Let $\Delta(z)$ be the perturbation determinant of A relative to A_0 , normalized at infinity.

Definition 5.1 (Perturbation determinant). For $z \in \mathbb{C} \setminus \mathbb{R}$, define

$$\Delta(z) := \det \left(\text{Id} + K(A_0 - z)^{-1} \right),$$

with canonical analytic continuation to an entire function determined by the growth estimates in Appendix C.

Proposition 5.2 (Functional symmetry and growth). $\Delta(z)$ is entire, of finite order and type determined by $\widehat{\kappa_{\phi_g}}$

$$\overline{\Delta(\bar{z})} = \Delta(z), \quad \Delta(-z) = \Delta(z) e^{p(z)}$$

for some even polynomial p of degree ≤ 2 .

Proof. Evenness and reality of the symbol imply $\overline{\Delta(\bar{z})} = \Delta(z)$. The rational decay $(1 + \phi_g^2 \omega^2)^{-1}$ fixes the genus;

Theorem 5.3 (Hadamard locking to Ξ). *There exist $C \neq 0$ and an even polynomial q of degree ≤ 2 such that*

$$\Delta(z) = C e^{q(z)} \Xi(z).$$

Proof. By Corollary 4.3, Δ and Ξ have matching zero-counting asymptotics. The trace identity (4.1) equates the line involution). The multiplicative ambiguity between two entire functions of the same order, with the same zero

Corollary 5.4 (Spectral bijection). *The zeros of Ξ coincide with the spectrum of A (counted with multiplicity) $\lambda = z$.*

Proof. Zeros of Δ are precisely the poles of the resolvent and hence eigenvalues of A . Apply Theorem 5.3. $\square$

6 de Branges positivity and critical-line localization

Define the scattering/transfer function

$$S(z) := \Delta(-z) \frac{1}{\Delta(z), \quad m(z) := i \frac{1+S(z)}{1-S(z)}}.$$

By Proposition 5.2, S is Schur-class in the upper half-plane and m is a Herglotz function.

Proposition 6.1 (Hermite–Biehler structure). *There exists a Hermite–Biehler entire function E such that*

$$S(z) = E^\#(z) \frac{1}{E(z), \quad \Xi(z) \propto E(z)E^\#(z)},$$

and the de Branges space $H(E)$ is well-defined with canonical reproducing kernel.

Proof. Schur/Herglotz factorization yields E with no zeros in $\text{Im } z > 0$, and $E^\#(z) := \overline{E(\bar{z})}$. The identity for Ξ follows.

Main Theorem 1 (Positivity forces the critical line). *All zeros of Ξ are real. Equivalently, thenontrivial zeros of ζ lie on the critical line $\frac{1}{2}$.*

Proof. In $H(E)$, the reproducing kernel positivity and the Hermite–Biehler property imply that the zeros of $E(z)E^\#(z)$ lie on the real axis. Since $\Xi(z) \propto E(z)E^\#(z)$, all zeros of Ξ are real. The equivalence with

Remark 6.2. The golden kernel fixes both the determinant’s growth (hence genus/type and the allowed polynomial q) and the Schur/Herglotz class of S . This dual control welds arithmetic (trace) and analysis (positivity) into a single invariant choice, removing tuning freedom.

7 Conclusion

We presented a canonical operator framework anchored by a golden-ratio resonance kernel, delivering (i) an explicit-trace identity, (ii) a determinant identity $\Delta = Ce^q \Xi$ pinning the spectrum to ζ ’s zeros, and (iii) line localization. The architecture minimizes arbitrariness: the single geometric choice ϕ_g dictates decay, growth, and

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A Kernel estimates and bounded multipliers

Lemma A.1. *Let $\widehat{\kappa_{\phi_g}}(\omega) = (1 + \phi_g^2 \omega^2)^{-1}$. For any even cutoff $f\eta \in C_c^\infty(\mathbb{R})$ and $X \geq 1$, the symbol S_X in (3.1) is real and uniformly in X .*

Idea of proof. Compact support of $u \mapsto \eta(u/X)$ restricts $\log n$ to a bounded window. In that window, the weights

B The explicit formula via trace identities

We outline the standard derivation of the explicit formula by comparing $\text{Tr}(\varphi(A))$ against $\text{Tr}(\varphi(A_0))$ for even φ . A side computation reduces $\text{Tr}(\varphi(A_X) - \varphi(A_0))$ to integrating $\widehat{\varphi}$ against S_X . As $X \rightarrow \infty$, the prime sum term matches resolvent convergence (Proposition 3.2) and dominated convergence.

C Determinants, Hadamard factorization, and growth

For bounded self-adjoint K , the operator $\text{Id} + K(A_0 - z)^{-1}$ is a trace-class perturbation of the identity for $z \notin \mathbb{R}$. The perturbation determinant $\Delta(z)$ is analytic on $\mathbb{C} \setminus \mathbb{R}$ and extends to an entire function controlled by the decay of $\Delta(z)e^{p(z)}$ stems from evenness of the kernel; the polynomial p is constrained by the order/type dictated by $\widehat{\kappa_{\phi_g}}$. Hence $Ce^q \Xi$, with q even of degree ≤ 2 . Normalization at infinity fixes C, q .

D Scattering, spectral shift, and Birman–Kreĭn

The spectral shift function $\xi(\lambda)$ for the pair (A, A_0) satisfies

$$\det \left(\text{Id} + K(A_0 - (\lambda + i0))^{-1} \right) = \exp \left(-i\pi \xi(\lambda) \right).$$

Coupling this with the trace identity in Theorem 4.2 relates $\xi'(\lambda)$ to the explicit formula density, which in turn

E de Branges spaces and positivity

Given a Schur function S in $\text{Im } z > 0$, the Herglotz transform $m(z) = i \frac{1+S}{1-S}$ admits a canonical representation. – Biehl entire E with $S = E^\# / E$. The de Branges space $\mathcal{H}(E)$ consists of entire F with

$$\|F\|_{\mathcal{H}(E)}^2 = \int_{\mathbb{R}} \frac{|F(x)|^2}{|E(x)|^2} dx < \infty,$$

and has reproducing kernel

$$K(z, w) = E(z)E^\#(\bar{w}) - E^\#(z)E(\bar{w}) \overline{2\pi i(\bar{w} - z)}.$$

Positivity of K forces zeros of $EE^\#$ to lie on $\mathbb{R}$, implying Main Theorem 1 once $\Xi \propto EE^\#$.

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